whence $\cos^2\alpha + \frac{y'^2}{b^2} = 1$

$\Rightarrow$ $y' = b\sin\alpha$.

Hence P is $(a\cos\alpha, b\sin\alpha)$.

The parameter α is called the eccentric angle of the point P . Note that α is the angle which the line CQ makes with the x -axis and not the angle which the line CP makes with it.

Hence the parametric equation of an ellipse is $x = a\cos\theta$, $y = b\sin\theta$, where θ is the parameter $0 \le \theta \le 2\pi$.

(iv) Parametric form of the Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

Similarly, parametric equation of a hyperbola can be derived as $x = a\sec\theta$, $y = b\tan\theta$, where θ is the parameter. $-\pi \le \theta \le \pi$ except $\theta = \pm\frac{\pi}{2}$.

In nutshell the parametric equations of the circle, parabola,ellipse and hyperbola are given in the following table.

Conic	Parametric equations	Parameter	Range of parameter	Any point on the conic
Circle	$x = a\cos\theta$ $y = a\sin\theta$	θ	$0 \le \theta \le 2\pi$	'θ' or $(a\cos\theta, a\sin\theta)$
Parabola	$x = at^2$ $y = 2at$	t	$-\infty < t < \infty$	't' or $(at^2, 2at)$
Ellipse	$x = a\cos\theta$ $y = b\sin\theta$	θ	$0 \le \theta \le 2\pi$	'θ' or $(a\cos\theta, b\sin\theta)$
Hyperbola	$x = a\sec\theta$ $y = b\tan\theta$	θ	$-\pi \le \theta \le \pi$ except $\theta = \pm\frac{\pi}{2}$	'θ' or $(a\sec\theta, b\tan\theta)$

Remark

(1) Parametric form represents a family of points on the conic which is the role of a parameter. Further parameter plays the role of a constant and a variable, while cartesian form represents the locus of a point describing the conic. Parameterisation denotes the orientation of the curve.

(2) A parametric representation need not be unique.

(3) Note that using parameterisation reduces the number of variables at least by one.

5.6 Tangents and Normals to Conics

Tangent to a plane curve is a straight line touching the curve at exactly one point and a straight line perpendicular to the tangent and passing through the point of contact is called the normal at that point.

5.6.1 Equation of tangent and normal to the parabola $y^2 = 4ax$

(i) Equation of tangent in cartesian form

Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ be two points on a parabola $y^2 = 4ax$.

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